

Aryan Mondal

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EDUCATION

University of Massachusetts Amherst

MS in Computer Science (Incoming)

Amherst, MA

Sep 2026 – May 2028

SRM Institute of Science and Technology

B.Tech Computer Science and Engineering, CGPA: 8.39/10

Chennai, Tamil Nadu

Sep 2022 – May 2026

- Relevant Coursework: Cryptography and Network Security, Forensics and Incident Response, Security Management, Computer Networks, Cyberwarfare, Mobile and Wireless Security, Hacker Techniques, Tools, and Incident Handling

EXPERIENCE

Security Intern

Thirumoolar IT Solutions

Feb 2025 – Mar 2025

Chennai, Tamil Nadu

- Implemented 20+ vulnerability scans and penetration tests using OWASP ZAP, Burp Suite, and Nmap, identifying 40+ security issues and contributing to a 25% improvement in system security posture.
- Investigated codebases of 3 internal web applications by applying manual code analysis, detecting SQL Injection and XSS vulnerabilities and enabling developers to resolve a significant share of exploitable security flaws.

PUBLICATIONS

An Adaptive Feedback Architecture for Stabilizing SNSPDs in TF-QKD Networks using SDQN

IEEE Transactions on Quantum Engineering (TQE)

Under Review

- Engineered an SDQN with hot-swappable PID and Bayesian optimization controllers by decoupling the classical control plane from the quantum data plane, achieving a 41% Secret Key Rate improvement and $4\times$ reduction in SKR variance (σ_{SKR} : 0.180 $\rightarrow$ 0.038).
- Modeled SNSPD photon detection efficiency using Ginzburg–Landau critical current equations by deploying a Matérn 5/2 Gaussian Process with Expected Improvement acquisition, converging to 95% of the optimal operating point in 80 cycles versus 120 for PID alone.
- Implemented Mahalanobis Distance anomaly detection across correlated SKR, QBER, temperature, and bias metrics by triggering an automated Security Mode clamp upon a 3σ breach, defeating blinding and thermal manipulation attacks.

A Robust and Lightweight Authentication Protocol for UAV Swarm Communications using

Jul 2026

HKDF-based Key Derivation and AEAD

IEEE CONECCT 2026

- Designed R-HAAP by replacing a Duffing chaotic-map keystream with a verified ECDH $\rightarrow$ HKDF $\rightarrow$ ChaCha20-Poly1305 pipeline, achieving 150–250 μs per-packet latency on ARM Cortex-M4 — 20–33 $\times$ faster than the original protocol and immune to cache-timing side-channel attacks.
- Implemented per-packet key derivation by constructing a unique HKDF.expand info string from UAV ID and monotonic offset counter, cryptographically binding metadata to the Poly1305 MAC and providing provable replay and forgery resistance.
- Validated swarm scalability using UavNetSim-v1 by simulating star and FANET mesh topologies across 10–100 UAVs, demonstrating 99.8%+ packet delivery and under 15 ms end-to-end latency within the 100 ms real-time flight control threshold.

Lightweight Hybrid DNN-GNN Architecture for Network Intrusion Detection with Adaptive Late Fusion

Apr 2026

ICNWC 2026

- Designed a hybrid DNN ($\sim 68\text{K}$ params) and Pseudo-GNN ($\sim 8\text{K}$ params) with adaptive late fusion by combining parallel statistical and relational branches, achieving 99.64% accuracy on 60K CIC-IDS-2017 samples at $1,250\times$ fewer parameters than Transformer-based NIDS.
- Analyzed a learnable fusion weight α by tracking its convergence from 0.689 to 0.389 over 25 epochs, establishing relational features as more discriminative and confirming the 8K pseudo-GNN branch outperforms the 68K DNN-only baseline.
- Benchmarked the model against ConvNeXt-Tiny (28M params) and a Transformer NIDS (100M params) by evaluating on 9,000 held-out samples, achieving competitive accuracy at $350\times$ and $1,250\times$ fewer parameters with a 0.08% false positive rate.

Quantum Key Distribution for SCADA in Power Grids: Security Architectures and Practical Applications

Nov 2025

ICETCE 2025

- Developed SVM and Random Forest classifiers for QKD-enhanced SCADA anomaly detection by training on 10,000 simulated data points via GridSearchCV, achieving 94.9% accuracy with SVM attaining 100% recall.
- Evaluated four classifiers — SVM, Random Forest, KNN, and Neural Network — by comparing precision (0.68–0.70), recall (0.94–1.00), and F1-score (0.80–0.81), identifying SVM as optimal and exposing KNN's complete failure under class imbalance.
- Established a replicable QKD-SCADA security framework by researching 30+ field deployments across Tokyo, Vienna, and China's UHV networks, mapping BB84 and decoy-state QKD architectures to SCADA threat vectors including FDIA and MitM.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Bash, SQL

Cyber Security & Networking: Cryptography, Digital Forensics, Anomaly Detection, Malware Analysis, Penetration Testing, Web Exploits SQLi/XSS, Operating Systems, SIEM/SOC Operations, SCADA Security, UAV Security, Quantum Key Distribution QKD

Tools & Frameworks: Claude Code, OWASP ZAP, Burp Suite, Nmap, Wireshark, Scapy, PyShark, Apache Kafka, FastAPI, Grafana, TensorFlow, Keras

Other Skills: Risk Management, Compliance & Regulation, Security Awareness, Threat Intelligence, Strategic Planning, Project Management